

PostgreSQL Commands Cheat Sheet

psql meta-commands, connection strings, pg_dump, pg_restore, pg_ctl, config params.

psql — Connection

Connection String Formats

```
# URI format
psql "postgresql://user:password@host:5432/dbname?sslmode=require"
psql "postgres://user@host/db"

# Flag format
psql -h host -p 5432 -U user -d dbname -W # -W prompts for password

# Environment variables (no password in command)
export PGHOST=host PGPORT=5432 PGUSER=user PGDATABASE=db PGPASSWORD=pass
psql

# .pgpass file (auto password, no prompt)
# Format: host:port:dbname:user:password
echo "localhost:5432:mydb:myuser:mypass" >> ~/.pgpass
chmod 600 ~/.pgpass
```

Common psql Flags

Flag	Purpose
-h host	Host (default: localhost via UNIX socket)
-p port	Port (default: 5432)
-U user	Username
-d dbname	Database name
-W	Force password prompt
-w	Never prompt for password
-c 'SQL'	Run SQL and exit
-f file.sql	Run SQL file
-o output.txt	Write output to file
-A	Unaligned output (good for CSV-like)
-F ',','	Field separator (with -A)
-t	Tuples only (no column headers)
-q	Quiet mode (suppress startup message)

<code>-v var=val</code>	Set psql variable
<code>--echo-all</code>	Print all SQL commands

```
# Run query and export as CSV
psql -h host -U user -d db -A -F ',' -c "SELECT * FROM orders" > output.csv

# Run a file
psql -h host -U user -d db -f migration.sql

# Run inline SQL
psql -h host -U user -d db -c "SELECT version();"

```

psql Meta-Commands (Backslash Commands)

Connection & Session

Command	Description
<code>\c dbname [user]</code>	Connect to database
<code>\c -</code>	Reconnect (same settings)
<code>\conninfo</code>	Show current connection info
<code>\q</code>	Quit psql
<code>\!</code>	Shell command
<code>\! ls</code>	Execute shell command
<code>\set VAR value</code>	Set psql variable
<code>\echo :VAR</code>	Print variable
<code>\timing [on off]</code>	
<code>\pset pager off</code>	Disable pager (less)
<code>\x [on off]</code>	
<code>\pset format unaligned</code>	Unaligned output
<code>\pset fieldsep ','</code>	Set field separator
<code>\o filename</code>	Redirect output to file
<code>\o</code>	Stop redirecting output
<code>\i filename</code>	Execute SQL file
<code>\ir filename</code>	Execute relative to current file
<code>\e</code>	Edit query in \$EDITOR

Information / Describe Commands

Command	Description
\l or \list	List databases
\l+	List databases with size
\dn	List schemas
\dn+	List schemas with permissions
\dt	List tables (current schema)
\dt *,*	List ALL tables in ALL schemas
\dt schema.*	List tables in specific schema
\dt+ tablename	Describe table with size/description
\d tablename	Describe table (columns, indexes, FKs)
\d+ tablename	Describe table (verbose, storage params)
\di	List indexes
\di+ indexname	Index details with size
\ds	List sequences
\dv	List views
\dv+ viewname	View definition
\dm	List materialized views
\df	List functions
\df funcname	Function details
\df+	Functions with source
\dp tablename	Table permissions (ACL)
\du	List roles/users
\du rolename	Role details
\dg	List groups
\dS	List system catalog tables
\dT	List data types
\dx	List installed extensions
\db	List tablespaces
\dC	List casts

<code>\do</code>	List operators
<code>\dF</code>	List text search configurations
<code>\dFd</code>	List text search dictionaries
<code>\sf funcname</code>	Show function source
<code>\sv viewname</code>	Show view source

Formatting

```

\x          -- toggle expanded view (very useful for wide tables)
\x auto     -- auto-switch based on terminal width
\pset null 'NULL' -- show NULLs as 'NULL' instead of blank
\pset border 2 -- enhanced borders (0=none, 1=plain, 2=full)
\a          -- toggle aligned/unaligned
\H          -- toggle HTML output

```

History & Editing

```

\l          -- list recent commands
\g          -- repeat last query
\g filename -- send last query output to file
\p          -- print query buffer
\r          -- reset query buffer
\s          -- command history
\s filename -- save history to file

```

System Views — Quick Reference

Connection / Activity

```

SELECT * FROM pg_stat_activity;           -- active connections
SELECT pid, username, state, query FROM pg_stat_activity WHERE state != 'idle';
SELECT count(*) FROM pg_stat_activity;    -- connection count
SELECT pg_terminate_backend(pid);        -- kill connection
SELECT pg_cancel_backend(pid);           -- cancel query (softer)

```

Tables & Indexes

```

SELECT * FROM pg_stat_user_tables;        -- table stats (seq scans, row counts,
vacuum)
SELECT * FROM pg_stat_user_indexes;       -- index usage
SELECT * FROM pg_statio_user_tables;     -- I/O stats
SELECT * FROM pg_indexes WHERE tablename = 't'; -- indexes on table
SELECT * FROM pg_tables WHERE schemaname = 'public'; -- tables

```

Locks

```
SELECT * FROM pg_locks;                -- all locks
SELECT * FROM pg_blocking_pids(pid);    -- what's blocking a pid

-- Find lock contention
SELECT
    blocked_locks.pid AS blocked_pid,
    blocking_locks.pid AS blocking_pid,
    blocked_activity.query AS blocked_query,
    blocking_activity.query AS blocking_query
FROM pg_catalog.pg_locks blocked_locks
JOIN pg_catalog.pg_stat_activity blocked_activity ON blocked_activity.pid =
blocked_locks.pid
JOIN pg_catalog.pg_locks blocking_locks ON blocking_locks.locktype =
blocked_locks.locktype
    AND blocking_locks.pid != blocked_locks.pid
    AND blocking_locks.granted
JOIN pg_catalog.pg_stat_activity blocking_activity ON blocking_activity.pid =
blocking_locks.pid
WHERE NOT blocked_locks.granted;
```

Performance

```
SELECT * FROM pg_stat_statements ORDER BY total_exec_time DESC LIMIT 20;  -- slow
queries
SELECT * FROM pg_stat_bgwriter;                                           --
checkpoint stats
SELECT * FROM pg_stat_replication;                                       --
replication lag
SELECT now() - pg_last_xact_replay_timestamp() AS replication_lag;       -- lag on
replica
```

Size Queries

```
-- Database sizes
SELECT datname, pg_size_pretty(pg_database_size(datname)) FROM pg_database;

-- Table sizes
SELECT tablename, pg_size_pretty(pg_total_relation_size(tablename::regclass)) AS
total_size,
    pg_size_pretty(pg_relation_size(tablename::regclass)) AS table_size,
    pg_size_pretty(pg_indexes_size(tablename::regclass)) AS index_size
FROM pg_tables WHERE schemaname = 'public' ORDER BY
pg_total_relation_size(tablename::regclass) DESC;

-- Index sizes
SELECT indexname, pg_size_pretty(pg_relation_size(indexname::regclass)) AS size
FROM pg_indexes WHERE schemaname = 'public' ORDER BY
```

```
pg_relation_size(indexname::regclass) DESC;

-- Unused indexes
SELECT idx.indexname, idx.tablename, i.idx_scan
FROM pg_stat_user_indexes i
JOIN pg_indexes idx ON i.indexrelname = idx.indexname
WHERE i.idx_scan = 0 AND NOT idx.indisunique;
```

pg_dump / pg_restore

pg_dump — Backup

```
# Logical dump (plain SQL)
pg_dump -h host -U user -d dbname -f backup.sql

# Custom format (best for pg_restore, supports parallel restore)
pg_dump -h host -U user -d dbname -Fc -f backup.dump

# Directory format (parallel dump)
pg_dump -h host -U user -d dbname -Fd -j 4 -f backup_dir/

# Compress output
pg_dump -h host -U user -d dbname -Fc -Z 9 -f backup_compressed.dump

# Table-level backup
pg_dump -h host -U user -d dbname -t tablename -f table_backup.sql

# Schema only (no data)
pg_dump -h host -U user -d dbname -s -f schema_only.sql

# Data only (no schema)
pg_dump -h host -U user -d dbname -a -f data_only.sql

# Exclude a table
pg_dump -h host -U user -d dbname -T exclude_table -f backup.sql

# Specific schema
pg_dump -h host -U user -d dbname -n schema_name -f schema_backup.sql
```

pg_dump Flags Reference

Flag	Description
-F c	Custom format (compressed)
-F d	Directory format
-F t	Tar format
-F p	Plain SQL (default)

-j n	Parallel workers (with -Fd)
-Z n	Compression level (0-9)
-s	Schema only
-a	Data only
-t table	Specific table
-T table	Exclude table
-n schema	Specific schema
-N schema	Exclude schema
--no-owner	No ownership commands
--no-privileges	No GRANT/REVOKE
--column-inserts	INSERT statements instead of COPY
--if-exists	Add IF EXISTS to DROP statements

pg_restore

```
# Restore from custom format
pg_restore -h host -U user -d dbname backup.dump

# Restore with parallel workers
pg_restore -h host -U user -d dbname -j 4 backup.dump

# Restore only schema
pg_restore -h host -U user -d dbname -s backup.dump

# Restore only data
pg_restore -h host -U user -d dbname -a backup.dump

# Restore specific table
pg_restore -h host -U user -d dbname -t tablename backup.dump

# List contents of dump
pg_restore --list backup.dump

# Restore with verbose output
pg_restore -h host -U user -d dbname -v backup.dump

# Restore plain SQL dump
psql -h host -U user -d dbname -f backup.sql
```

pg_dumpall — Backup All Databases

```
pg_dumpall -h host -U postgres -f all_databases.sql # includes roles and
tablespaces
pg_dumpall -h host -U postgres -g -f roles.sql      # roles/globals only
```

pg_ctl — Server Control

```
# Start PostgreSQL
pg_ctl -D /var/lib/postgresql/data start
pg_ctl -D /var/lib/postgresql/data start -l logfile

# Stop (modes: smart=wait for clients, fast=disconnect clients, immediate=crash-
stop)
pg_ctl -D /var/lib/postgresql/data stop -m fast
pg_ctl -D /var/lib/postgresql/data stop -m smart
pg_ctl -D /var/lib/postgresql/data stop -m immediate # dangerous

# Restart
pg_ctl -D /var/lib/postgresql/data restart -m fast

# Reload config (no restart needed for most params)
pg_ctl -D /var/lib/postgresql/data reload
-- OR from psql:
SELECT pg_reload_conf();

# Status
pg_ctl -D /var/lib/postgresql/data status

# Initdb (initialize new cluster)
pg_ctl initdb -D /var/lib/postgresql/data

# Promote standby to primary
pg_ctl -D /var/lib/postgresql/data promote
```

systemd Alternative (common on Linux)

```
systemctl start postgresql
systemctl stop postgresql
systemctl restart postgresql
systemctl reload postgresql
systemctl status postgresql
```

psql — Useful Query Patterns

```
-- List all tables with row count estimate
SELECT schemaname, tablename, n_live_tup AS est_row_count
FROM pg_stat_user_tables ORDER BY n_live_tup DESC;
```



```

-- Find tables with no primary key
SELECT tablename FROM pg_tables t
WHERE schemaname = 'public'
    AND NOT EXISTS (SELECT 1 FROM pg_constraint c
                    JOIN pg_class r ON c.conrelid = r.oid
                    WHERE r.relname = t.tablename AND c.contype = 'p');

-- Find long-running queries
SELECT pid, now() - query_start AS duration, query
FROM pg_stat_activity
WHERE state = 'active'
    AND query_start < now() - INTERVAL '5 minutes'
ORDER BY duration DESC;

-- Vacuum and analyze stats
SELECT relname, last_vacuum, last_autovacuum, last_analyze, last_autoanalyze,
n_dead_tup
FROM pg_stat_user_tables
ORDER BY n_dead_tup DESC;

-- Bloat estimate
SELECT tablename,
       n_dead_tup,
       n_live_tup,
       ROUND(100.0 * n_dead_tup / NULLIF(n_live_tup + n_dead_tup, 0), 2) AS dead_pct
FROM pg_stat_user_tables
WHERE n_dead_tup > 10000
ORDER BY dead_pct DESC;

```

VACUUM and ANALYZE

```

VACUUM tablename;           -- reclaim dead tuples
VACUUM FULL tablename;      -- reclaim space (locks table, rewrites)
VACUUM ANALYZE tablename;   -- vacuum + update stats
VACUUM VERBOSE tablename;   -- with progress output
ANALYZE tablename;          -- update planner statistics only
ANALYZE tablename (column1, col2); -- stats for specific columns

-- Check autovacuum settings
SHOW autovacuum;
SHOW autovacuum_vacuum_threshold;
SHOW autovacuum_vacuum_scale_factor;

-- Disable autovacuum for a table (NOT recommended for production)
ALTER TABLE mytable SET (autovacuum_enabled = false);

```

PostgreSQL Config Parameter Reference (postgresql.conf)

Memory

Parameter	Default	Notes
shared_buffers	128MB	Set to 25% of RAM
effective_cache_size	4GB	Set to 75% of RAM (hint for planner)
work_mem	4MB	Per sort/hash operation. Multiply by max_connections × 2 for total risk
maintenance_work_mem	64MB	For VACUUM, CREATE INDEX. Set to 256MB-1GB
wal_buffers	-1 (auto)	Default is 1/32 of shared_buffers, max 64MB
temp_buffers	8MB	Per session temp tables

Write-Ahead Log

Parameter	Default	Notes
wal_level	replica	Options: minimal, replica, logical
synchronous_commit	on	Set off for async (10x faster writes, lose last 200ms on crash)
checkpoint_completion_target	0.9	Spread checkpoint I/O
checkpoint_timeout	5min	Max time between checkpoints
max_wal_size	1GB	WAL size before forcing checkpoint
min_wal_size	80MB	Minimum WAL retention
wal_compression	off	Enable for network-heavy replication

Connection

Parameter	Default	Notes
max_connections	100	Each connection ~5-10MB RAM. Use PgBouncer
superuser_reserved_connections	3	Reserved for superuser login when pool is full
idle_in_transaction_session_timeout	0 (off)	Set to 60000 (1 min) to kill idle-in-txn
statement_timeout	0 (off)	Kill long-running queries automatically
lock_timeout	0 (off)	Fail if can't acquire lock within N ms

Planner

Parameter	Default	Notes
random_page_cost	4.0	Set to 1.1 for SSDs

seq_page_cost	1.0	Leave at default
effective_io_concurrency	1	Set to 200 for SSDs
default_statistics_target	100	Increase to 200–500 for better estimates on complex queries
enable_parallel_query	on	Enable parallel scans
max_parallel_workers_per_gather	2	Set to num_CPU_cores / 2
max_parallel_workers	8	Limit total parallel workers

Autovacuum

Parameter	Default	Notes
autovacuum	on	Never turn off
autovacuum_vacuum_threshold	50	Min dead rows before vacuum
autovacuum_vacuum_scale_factor	0.2	Vacuum when 20% of table is dead
autovacuum_analyze_scale_factor	0.1	Analyze when 10% of table changed
autovacuum_vacuum_cost_delay	2ms	Throttle autovacuum I/O impact
autovacuum_max_workers	3	Increase for write-heavy workloads

Logging

```
log_min_duration_statement = 1000    # Log queries > 1 second
log_checkpoints = on
log_connections = off
log_disconnections = off
log_lock_waits = on
log_temp_files = 0                  # Log all temp files (0 = all)
log_autovacuum_min_duration = 250ms  # Log slow autovacuum
log_line_prefix = '%t [%p]: [%l-1] user=%u,db=%d,app=%a,client=%h '
```

Reload vs. Restart Requirements

Action	Required
pg_reload_conf() / pg_ctl reload	Most config changes (work_mem, logging, autovacuum settings)
Server restart	shared_buffers, max_connections, wal_level, listen_addresses
Super-user SET	Session-level: SET work_mem = '256MB';
ALTER SYSTEM	Writes to postgresql.auto.conf, survives restart

```
-- Check which parameters require restart
SELECT name, setting, unit, context FROM pg_settings WHERE pending_restart = TRUE;

-- Change config safely
ALTER SYSTEM SET work_mem = '64MB';
SELECT pg_reload_conf(); -- or restart if required
```

Extensions — Install and Use

```
-- List available extensions
SELECT * FROM pg_available_extensions ORDER BY name;

-- List installed extensions
SELECT * FROM pg_extension;
-- Or: \dx in psql

-- Install
CREATE EXTENSION IF NOT EXISTS pg_trgm;           -- trigram similarity
CREATE EXTENSION IF NOT EXISTS postgis;           -- geospatial
CREATE EXTENSION IF NOT EXISTS pgcrypto;          -- encryption
CREATE EXTENSION IF NOT EXISTS uuid-ossf;         -- UUID generation
CREATE EXTENSION IF NOT EXISTS pg_stat_statements; -- query stats
CREATE EXTENSION IF NOT EXISTS btree_gist;         -- GiST for scalar types
CREATE EXTENSION IF NOT EXISTS ltree;             -- hierarchical labels
CREATE EXTENSION IF NOT EXISTS hstore;            -- key-value pairs
CREATE EXTENSION IF NOT EXISTS tablefunc;         -- crosstab (pivot)
CREATE EXTENSION IF NOT EXISTS timescaledb;       -- time-series
CREATE EXTENSION IF NOT EXISTS citus;             -- horizontal sharding

-- Drop
DROP EXTENSION IF EXISTS extension_name;
```

User / Role Management

```
-- Create user
CREATE USER username WITH PASSWORD 'password';
CREATE USER username WITH PASSWORD 'password' CREATEDB;

-- Create role
CREATE ROLE rolename WITH LOGIN PASSWORD 'password';
CREATE ROLE readonly_role;

-- Grant permissions
GRANT SELECT ON ALL TABLES IN SCHEMA public TO readonly_role;
GRANT SELECT, INSERT, UPDATE ON TABLE orders TO app_user;
GRANT ALL PRIVILEGES ON DATABASE dbname TO admin_user;
GRANT EXECUTE ON FUNCTION func_name TO user;
```

```
GRANT CONNECT ON DATABASE dbname TO user;
GRANT USAGE ON SCHEMA schema TO user;

-- Default privileges (for future tables)
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT SELECT ON TABLES TO readonly_role;

-- Assign role to user
GRANT readonly_role TO username;

-- Revoke
REVOKE SELECT ON TABLE orders FROM user;

-- Drop user
DROP USER username;
DROP ROLE rolename;

-- Change password
ALTER USER username WITH PASSWORD 'newpassword';

-- List roles
\du -- in psql
SELECT * FROM pg_roles;
```